Image Colorization Using GANs

A U-Net Generator with a PatchGAN Discriminator

This repository implements a **deep learning–based image colorization system** using a **Conditional Generative Adversarial Network (cGAN)**.
The model learns to predict the color channels (a/b) of an image from its luminance (L-channel) using a U-Net generator and a PatchGAN discriminator.
The pipeline supports **training, validation, visualization, and inference on external images**.

🔍 Overview

- **Model Architecture**
  - U-Net Generator
  - PatchGAN Discriminator
- Works in **LAB color space**
- Supports: training, visualization, inference, dataset preprocessing

🗀 Project Structure

```
Image_Colorization/
├── Gans/
│   ├── data.py
│   ├── model.py
│   ├── train.py
│   ├── utils.py
├── gui.py
├── main.py
└── README.md
```

🧠 Model Architecture

Generator (U-Net)

- Encoder-decoder with skip connections
- Outputs ab channels

Discriminator (PatchGAN)

- Classifies patches instead of whole images
- More stable GAN training

Loss Functions

- GAN loss
- L1 reconstruction loss

📦 Installation

```
pip install torch torchvision pillow matplotlib numpy opencv-python requests
```

🚀 Training

```
python Gans.main.py
```

🎨 Visualization

python main.py

✔ Requirements

- Python 3.7+
- PyTorch
- (Optional) CUDA GPU